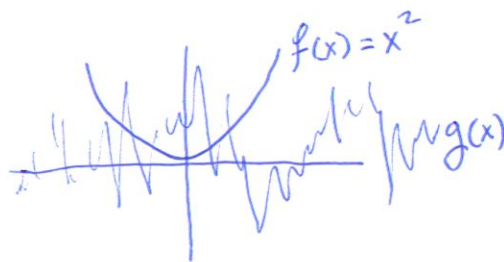


What is Analysis?

- Functions $f(x)$
- Inequalities $<, >, \leq, \geq$



What is it applied to?

- Differential equations (most of physics)
- Probability (statistics)
- Geometry

Some Goals

- ~~Basic topology~~ Basic topology

Integers $\mathbb{Z}$

.....

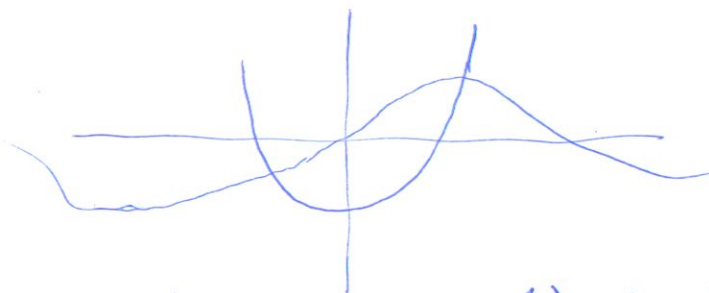
- non-compact
- discrete
- countable

Interval $[0, 1]$

—••

- compact
- connected
- uncountable

- Is there a solution to $1 - x^2 = \sin(x)$?



- Make calculus rigorous (à la Cauchy 1789-1857)

- What does $\frac{d}{dx} f(x)$ mean?
- What does $\int_a^b f(x) dx$ mean?
- Why ^a does $\frac{d}{db} \int_a^b f(x) dx = f(b)$?
(and when)

Sets

A set has elements

$x \in S$ "x is in S"

Example: Natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$

Example: Integers $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

Def If A and B are sets, and $x \in A$ implies $x \in B$, then A is a subset of B .



Note: $A \subset B$ and $B \subset A$ implies $A = B$

Technical note: all sets ~~are~~ considered are subsets of ~~the~~ a universe, which contains everything we might need.

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Def New sets from old. Let A, B be sets.

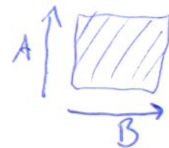
1) Union $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$



2) Intersection $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$



3) Product $A \times B = \{(x, y) \mid x \in A, y \in B\}$



4) Power Set $\mathcal{P}(A) = \{\text{subsets of } A\}$

~~###~~

Ex

$$\{1, 2\} \cup \{2, 3\} = \{1, 2, 3\}$$

$$\{1, 2\} \cap \{2, 3\} = \{2\}$$

$$\{1, 2\} \times \{2, 3\} = \{(1, 2), (1, 3), (2, 2), (2, 3)\}$$

$$\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

Functions

Def A, B sets. A function $f: A \rightarrow B$ is a rule associating to each $x \in A$ some $f(x) \in B$.

A is the domain and B is the codomain.

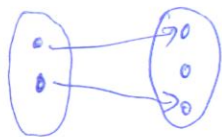
Ex $f(x) = \frac{1}{1+x^2}$ defines a function $\mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \frac{1}{1-x^2} \text{ defines a function } \mathbb{R} \setminus \{\pm 1\} \rightarrow \mathbb{R}.$$

4

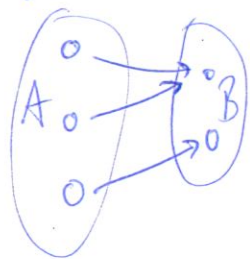
Ex "+" and "." are functions $\mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$.

Def $f: A \rightarrow B$ is injective if $f(x) = f(y)$ implies $x = y$.



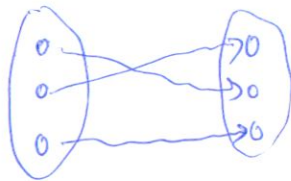
Q $f: \mathbb{Z} \rightarrow \mathbb{Z}$ $f(x) = x^2$ injective?
 $f: \mathbb{N} \rightarrow \mathbb{N}$ $f(x) = x^2$ injective?

Def $f: A \rightarrow B$ is surjective if $\forall b \in B, \exists a \in A$ s.t. $f(a) = b$.

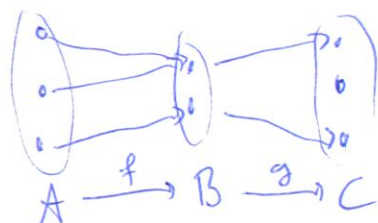


Q $f: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ $f(x, y) = xy$ surjective?
 $f: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ $f(x, y) = x^2 + y^2$ surjective?

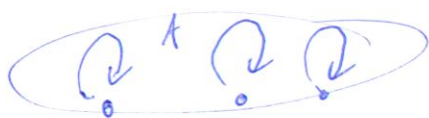
Def $f: A \rightarrow B$ is bijective if it is injective and surjective.



Def Given $f: A \rightarrow B$, $g: B \rightarrow C$, the composition $g \circ f: A \rightarrow C$ is defined by
 $(g \circ f)(a) = g(f(a))$.



Def Let A be a set. The identity $\text{Id}_A: A \rightarrow A$ is the function defined by $\text{Id}_A(x) = x \quad \forall x \in A$.



Prop $f: A \rightarrow B$, $g: B \rightarrow C$ functions.

1) f, g inj. $\Rightarrow g \circ f$ inj.

2) f, g surj. $\Rightarrow g \circ f$ surj.

3) f, g bij $\Rightarrow g \circ f$ bij. (follows from 1+2)

Proof of 1 Let $x, y \in A$.

Suppose $(g \circ f)(x) = (g \circ f)(y)$. WTS $x = y$.

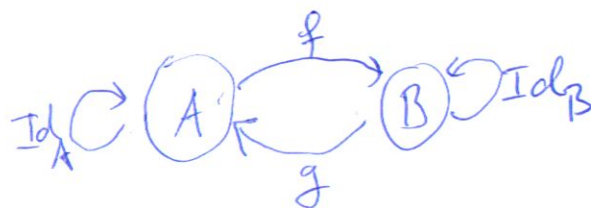
We have $g(f(x)) = g(f(y))$ by def. of composition.

We have $f(x) = f(y)$ because g is surjective.

Finally, $x = y$ because f is surjective.

Prop $f: A \rightarrow B$ is bij. iff it has a two-sided inverse.

That is, $g: B \rightarrow A$ s.t. $g \circ f = \text{Id}_A$ and $f \circ g = \text{Id}_B$.



Proof $\Leftarrow$ Suppose g is a two sided inverse.

First we show f is injective. Suppose $f(x) = f(y)$ for $x, y \in A$.

$g(f(x)) = g(f(y))$, so $x = y$.

f is surjective: Suppose $x \in B$. Consider $g(x) \in A$.

$f(g(x)) = x$.

$\Rightarrow$ Suppose $f: A \rightarrow B$ is bijective. For each $y \in B$, there is a unique $x \in A$ s.t. $f(x) = y$. It exists because f surj. and is unique because f inj. (If $f(x) = f(x') = y$ then $x = x'$). Define $g(y)$ to be this element of A for each $y \in B$.

Check g is two sided inverse: Suppose $y \in B$. 7

~~f~~ $f(g(y)) = y$ by definition of g . Suppose $x \in A$.

Want to show $g(f(x)) = x$. We know

$$f(g(f(x))) = f(x) \quad \text{because} \quad f \circ g = \text{Id}_B.$$

f is injective, so $g(f(x)) = x$. Thus, $g \circ f = \text{Id}_A$.